

Portfolio Risk & Return Analysis and Rebalancing Recommendation

Ultra-High-Net-Worth Client - Palo Alto, CA - \$95M Liquid Portfolio

SQL & Data Management - Individual Final Assessment

Analysis window: 2023-06-16 to 2026-06-15 - Risk-free rate: 3.68% (3M U.S. T-Bill)

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Executive Summary

The client holds a five-ETF, \$95M liquid portfolio. Recent performance has been strong (+29% over 12M), but it is driven almost entirely by two engines: U.S. mega-cap technology (IXN + QQQ) and gold (GLD). These two equity funds hold the same mega-cap names and are 0.95 correlated (BlackRock, 2026; Invesco, 2026b), so the portfolio is effectively reliant on a single market regime - the tech-and-gold bull - to keep performing. Modern portfolio theory shows that combining assets with lower correlations improves the risk-return trade-off (Markowitz, 1952).

Our recommendation keeps these two proven return engines but adds three securities that tend to perform in different regimes - global small-cap value (AVDV), inflation-protected Treasuries (VTIP) and diversified commodities (DBC) - all within the client existing 40 (equity)/30 (fixed income) /30 (real assets) risk profile. The optimized portfolio improves the Sharpe ratio from 1.37 to 1.53 and lowers 12-month volatility from 12.2% to 10.9%, while expected return is essentially unchanged.

Metric	Current	Optimized	Change
Annualized return (36M)	19.58%	19.11%	-0.47 pp
Volatility 12M	12.24%	10.89%	-1.35 pp
Volatility 6M	15.22%	13.59%	-1.63 pp
Sharpe ratio (36M)	1.367	1.531	+0.164

1. Returns (12M / 18M / 24M)

Holding	12M	18M	24M
IXN	66.69%	67.32%	74.16%
QQQ	39.92%	39.35%	55.07%
IEF	4.18%	6.53%	7.97%
VNQ	13.14%	10.94%	26.60%
GLD	27.19%	61.94%	84.78%
Portfolio	29.11%	37.56%	49.29%

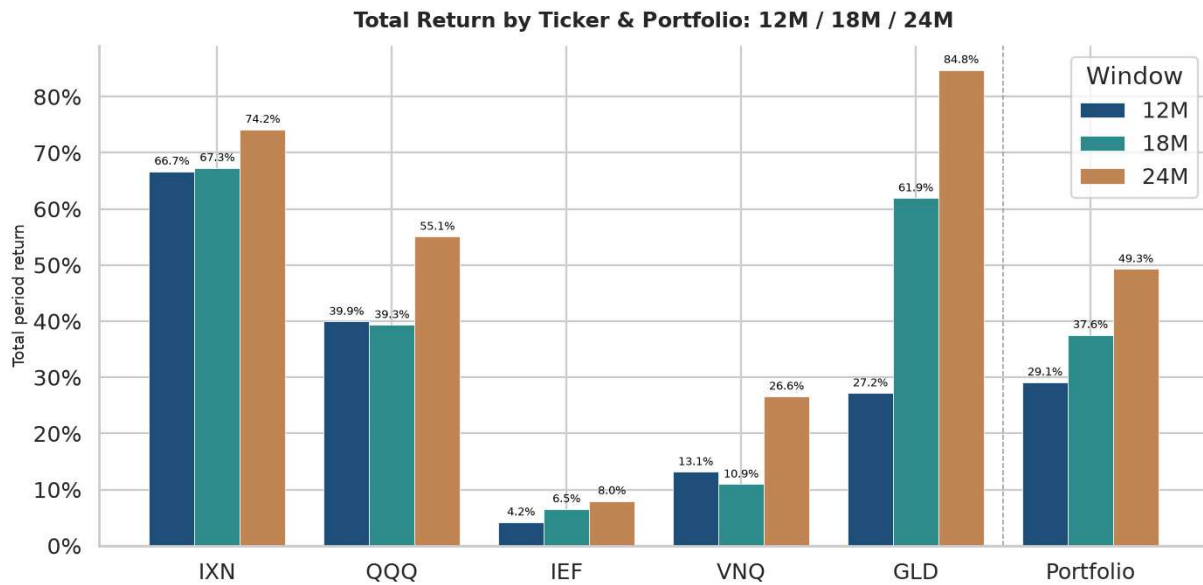


Figure 1 - Total returns by holding and for the whole portfolio.

Interpretation: the portfolio returned +29.1% (12M), +37.6% (18M) and +49.3% (24M). All of the return is generated by two engines - technology (IXN +66.7% / QQQ +39.9% over 12M) and gold (GLD +84.8% over 24M). The bond sleeve (IEF, +4.2% over 12M) was the laggard.

2. Correlations

Daily-return correlation matrix over the most-recent 12 and 6 months (Pearson; computed manually in MySQL because corr() is unavailable - see Appendix A).

12-Month correlation

	IXN	QQQ	IEF	VNQ	GLD
IXN	1.00	0.95	0.17	0.09	0.30
QQQ	0.95	1.00	0.18	0.17	0.30
IEF	0.17	0.18	1.00	0.41	0.24
VNQ	0.09	0.17	0.41	1.00	0.14
GLD	0.30	0.30	0.24	0.14	1.00

6-Month correlation

	IXN	QQQ	IEF	VNQ	GLD
IXN	1.00	0.96	0.37	0.12	0.38
QQQ	0.96	1.00	0.38	0.18	0.40
IEF	0.37	0.38	1.00	0.45	0.29
VNQ	0.12	0.18	0.45	1.00	0.14
GLD	0.38	0.40	0.29	0.14	1.00

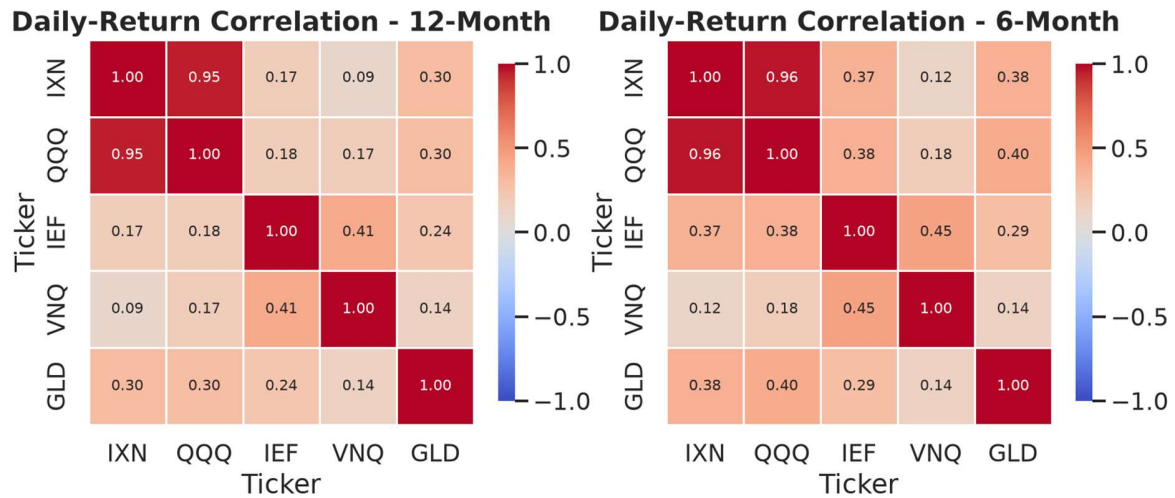


Figure 2 - Cross-asset correlation heatmaps (12M and 6M).

Interpretation: the strongest correlation is IXN-QQQ at 0.95 - effectively the same investment, yet together ~40% of the portfolio. The best diversifiers are gold and the bond/real-estate sleeve (VNQ-IXN as low as 0.09). In the 6-month window every correlation rises (e.g. IEF-equity 0.17 to 0.37, GLD-equity 0.30 to 0.38): diversification is weakening exactly when it is most needed.

3. Risk / Volatility (sigma)

Holding	Sigma 12M	Sigma 6M
IXN	24.14%	28.89%
QQQ	17.32%	19.72%
IEF	4.66%	4.82%
VNQ	13.48%	13.84%
GLD	27.34%	33.87%
Portfolio	12.24%	15.22%

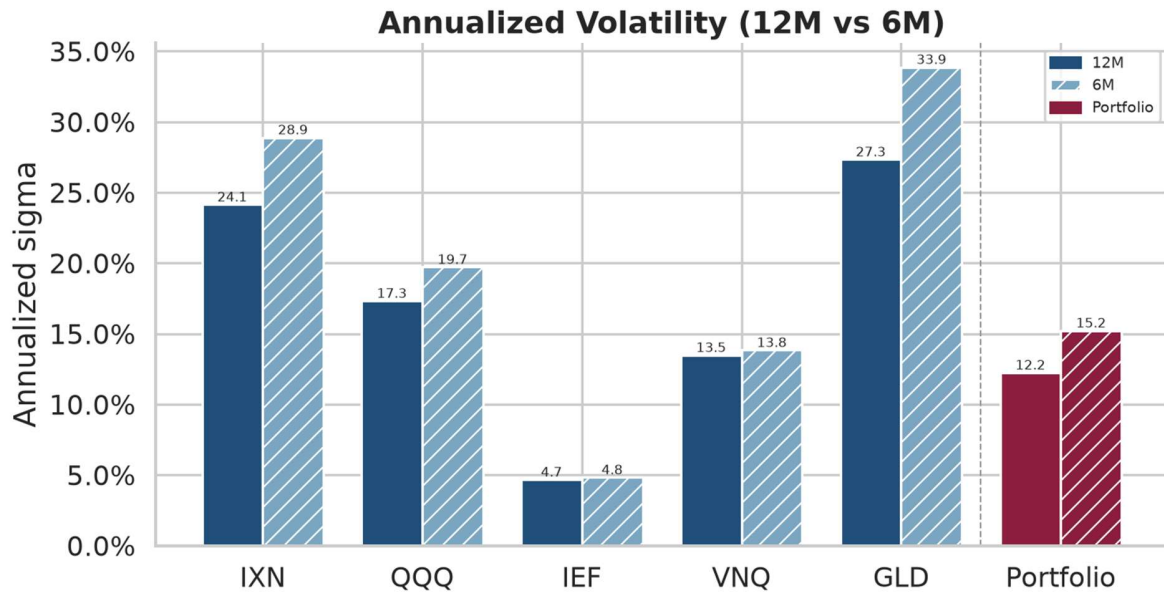


Figure 3 - Annualized volatility per holding and for the portfolio (12M vs 6M).

Interpretation: portfolio volatility is 12.24% (12M) rising to 15.22% (6M) - risk is accelerating. The largest risk contributor is gold (sigma 27% rising to 34%), followed by IXN (24% to 29%). IEF is the only true stabilizer (sigma ~4.7%).

4. Rebalancing Recommendation (Sell / Buy / Add)

We first compute the Sharpe ratio for every candidate security (procedure `q_sharpe`, 36M) to rank risk-adjusted performance; this drives the recommendation.

Asset	Ann. return	Ann. volatility	Sharpe
AVDV	25.21%	16.16%	1.332
GLD	28.11%	20.27%	1.205
IXN	31.93%	24.28%	1.164
QQQ	26.05%	19.94%	1.122
VTIP	5.08%	1.97%	0.713
DBC	11.90%	16.22%	0.507
VNQ	10.56%	16.77%	0.410
IEF	2.91%	6.54%	-0.118

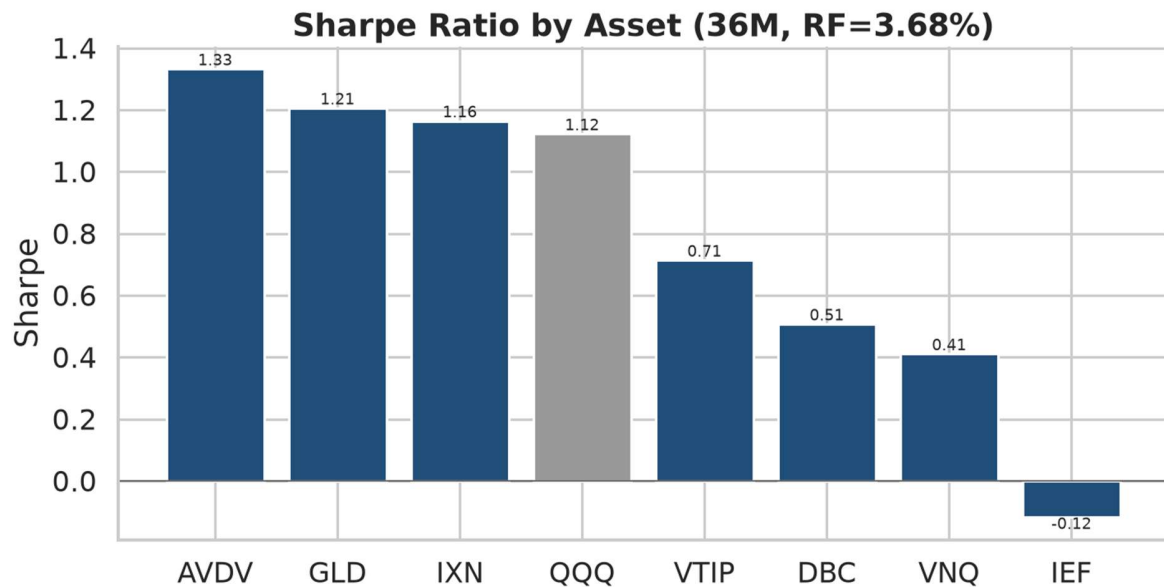


Figure 4 - Sharpe ratio by asset (36M). Blue = held in the optimized portfolio.

Qualitative rationale - why sell QQQ and keep IXN.

IXN and QQQ are near-substitutes: both are dominated by the same U.S. mega-cap technology stocks. Their published holdings confirm the overlap as of 31 March 2026 (BlackRock, 2026; Invesco, 2026b):

NVIDIA	12.2%	8.7%
Apple	10.8%	7.6%
Microsoft	7.4%	5.6%
Broadcom	4.7%	3.0%

Table - Shared top holdings of IXN and QQQ (BlackRock, 2026; Invesco, 2026b). IXN top-10 = ~57% of assets; QQQ top-10 = ~48%.

Given this redundancy (correlation 0.95), holding both adds risk concentration without diversification, so we remove one. We keep IXN rather than QQQ because IXN is a global technology fund: alongside the U.S. names it also holds non-U.S. leaders such as Taiwan Semiconductor (TSMC), giving the equity sleeve some geographic diversification that the U.S.-only Nasdaq-100 (QQQ) lacks (BlackRock, 2026; Invesco, 2026b).

Quantitative rationale - from the metrics above.

On a stand-alone basis IXN has both a higher Sharpe (1.16 vs 1.12) and a higher return than QQQ, so keeping IXN sacrifices nothing in risk-adjusted terms. IEF is the only holding with a negative Sharpe (-0.12) - it earns less than the risk-free rate - so it is cut. The additions are chosen on the same Sharpe evidence: AVDV has the best stand-alone Sharpe (1.33) and VTIP the best return-per-unit-of-risk through its very low volatility.

Sell / trim:

- **SELL QQQ entirely** - redundant with IXN (correlation 0.95) and a lower Sharpe.
- **CUT IEF (28.5% to 6%)** - the only holding with a negative Sharpe.

- **TRIM GLD (23% to 18%) and VNQ (8.9% to 5%)** - reduce single-commodity concentration and a low-Sharpe holding.

Keep (the two engines):

- **IXN (18%)** - global technology - primary growth engine; also adds non-US exposure (e.g., TSMC) (BlackRock, 2026).
- **GLD (18%)** - gold - store-of-value / safe-haven asset; only trimmed to control concentration (Baur & Lucey, 2010).

Buy / add (three outside securities for other regimes):

- **AVDV (22%)** - international small-cap value; captures the documented value premium and performs when value / non-US leads (Fama & French, 1993; Avantis Investors, 2026).
- **VTIP (24%)** - short-term TIPS; low volatility, inflation-protected, performs in rising-rate / inflation regimes (U.S. Department of the Treasury, n.d.; Vanguard, 2026).
- **DBC (7%)** - broad commodities (energy, metals, agriculture); an inflation hedge beyond a single metal (Invesco, 2026a).

5. Impact on Risk & Expected Return (Portfolio Sharpe comparison)

We compare the Sharpe ratio of the current vs optimized portfolio (procedures q_sharpe portfolio block and q5_optimized), together with returns and volatility - all computed in SQL.

Metric	Current	Optimized	Change
Annualized return (36M)	19.58%	19.11%	-0.47 pp
Volatility (36M)	11.63%	10.08%	-1.55 pp
Volatility 12M	12.24%	10.89%	-1.35 pp
Volatility 6M	15.22%	13.59%	-1.63 pp
Sharpe ratio (36M)	1.367	1.531	+0.164

Trade plan (AUM \$95M):

Holding	Current %	Opt %	Current \$	Opt \$	Trade \$M	Action
IXN	17.5%	18.0%	\$16.62M	\$17.10M	+0.48	BUY
QQQ	22.1%	0.0%	\$21.00M	\$0.00M	-21.00	SELL
AVDV	0.0%	22.0%	\$0.00M	\$20.90M	+20.90	BUY
IEF	28.5%	6.0%	\$27.07M	\$5.70M	-21.37	SELL
VTIP	0.0%	24.0%	\$0.00M	\$22.80M	+22.80	BUY
GLD	23.0%	18.0%	\$21.85M	\$17.10M	-4.75	SELL
DBC	0.0%	7.0%	\$0.00M	\$6.65M	+6.65	BUY
VNQ	8.9%	5.0%	\$8.46M	\$4.75M	-3.70	SELL

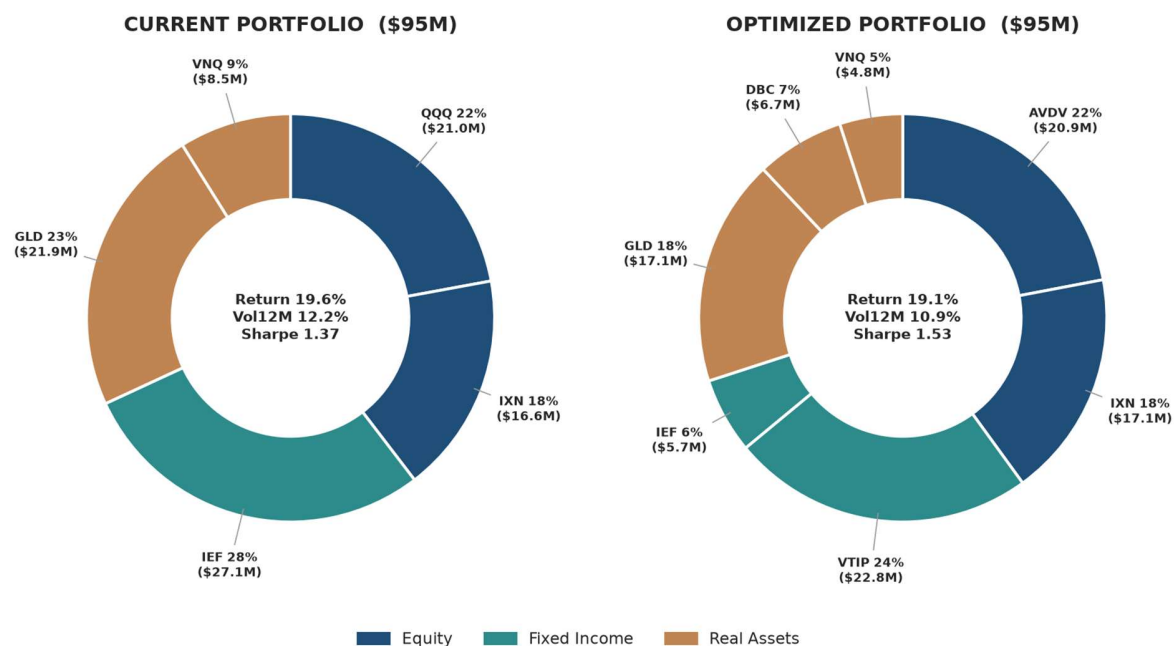


Figure 5 - Current vs Optimized allocation (same 40/30/30 risk profile).

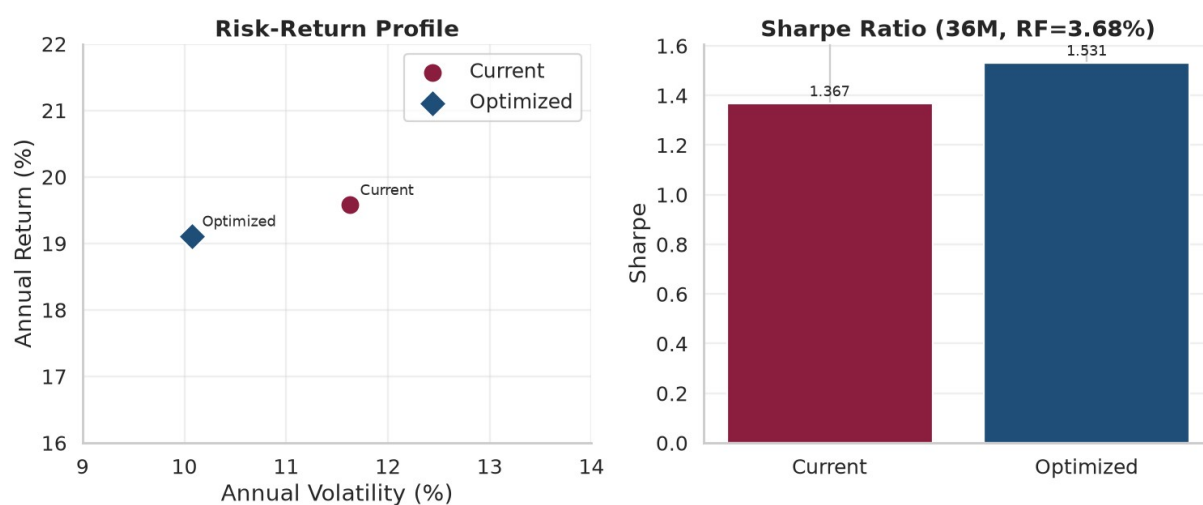


Figure 6 - Risk-return profile and portfolio Sharpe: Current vs Optimized.

Interpretation: the optimized portfolio raises the portfolio Sharpe ratio from 1.37 to 1.53 (+0.16) and lowers 12-month volatility by 1.3 percentage points, while expected return is essentially unchanged. By keeping technology and gold while adding value, TIPS and commodities, the client retains the upside engines but gains protection across a wider range of market regimes, at the same risk-profile budget.

6. Conclusion

The current portfolio is a strong but fragile performer: its high Sharpe ratio is the product of two correlated bets (U.S. tech and gold) that won together over this period. We keep those two engines but sell the redundant QQQ, cut the negative-Sharpe IEF, trim gold, and add three regime-diversifying securities (AVDV, VTIP, DBC) - all within the same 40/30/30 risk profile. The portfolio Sharpe ratio rises from 1.37 to 1.53 and volatility falls, without sacrificing expected return.

Data & Methodology

Analysis: every metric in this report is produced by a SQL stored procedure that takes the time window (months) as an input parameter (full code in Appendix A). The four metrics used are returns (Q1), correlations (Q2), volatility/sigma (Q3) and the Sharpe ratio current and optimized (Q4-Q5):

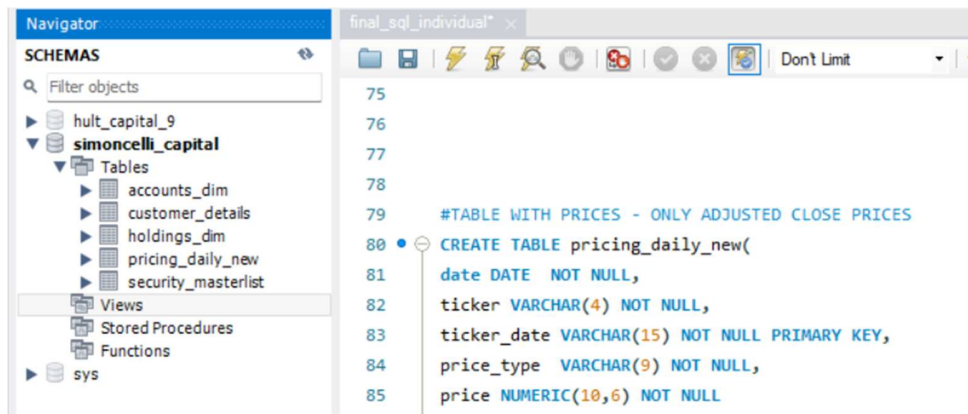
- returns are simple period returns (end price / start price - 1);
- Sharpe = (AVG(daily return) x 250 - 0.0368) / (STD(daily return) x sqrt(250)).); the 3.68% risk-free rate is the 3-month U.S. Treasury bill (Board of Governors of the Federal Reserve System, 2026).
- Returns are calculated using adjusted prices (total returns).
- Python was used as visualization tool.

References

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Appendix A - SQL Stored Procedures



```
DROP SCHEMA IF EXISTS Simoncelli_Capital;
```

```
CREATE SCHEMA Simoncelli_Capital;  
USE Simoncelli_Capital;
```

```
CREATE TABLE customer_details(  
customer_id INT,  
full_name VARCHAR(250),  
first_name VARCHAR(100),  
last_name VARCHAR(200),  
email VARCHAR(150),  
investor_category VARCHAR (100),  
location VARCHAR(250),  
PRIMARY KEY(customer_id)  
);
```

```
INSERT INTO customer_details(customer_id, full_name, first_name, last_name, email, investor_category, location) VALUES (1, 'Alexander Whitmore', 'Alexander', 'Whitmore', 'a.whitmore@email.com', 'Ultra High Net Worth', 'Palo Alto, CA');
```

```
INSERT INTO customer_details(customer_id, full_name, first_name, last_name, email, investor_category, location) VALUES (2, 'Victoria Chen', 'Victoria', 'Chen', 'v.chen@email.com', 'Ultra High Net Worth', 'Palo Alto, CA');
```

```
INSERT INTO customer_details(customer_id, full_name, first_name, last_name, email, investor_category, location) VALUES (3, 'Marcus Rothschild', 'Marcus', 'Rothschild', 'm.rothschild@email.com', 'Ultra High Net Worth', 'Palo Alto, CA');
```

```
INSERT INTO customer_details(customer_id, full_name, first_name, last_name, email, investor_category, location) VALUES (4, 'Isabella Navarro', 'Isabella', 'Navarro', 'i.navarro@email.com', 'Ultra High Net Worth', 'Palo Alto, CA');
```

```
CREATE TABLE accounts_dim(  
  account_id INT,  
  customer_id INT, #foreign key from customer_details  
  main_account INT,  
  acct_open_status INT,  
  acct_open_date VARCHAR(200),  
  PRIMARY KEY(account_id)  
);
```

```
INSERT INTO accounts_dim(account_id, customer_id, main_account, acct_open_status, acct_open_date) VALUES (101, 1, 1, 1, '2015-03-12');  
INSERT INTO accounts_dim(account_id, customer_id, main_account, acct_open_status, acct_open_date) VALUES (102, 1, 0, 1, '2017-08-25');  
INSERT INTO accounts_dim(account_id, customer_id, main_account, acct_open_status, acct_open_date) VALUES (103, 1, 0, 1, '2019-11-04');  
INSERT INTO accounts_dim(account_id, customer_id, main_account, acct_open_status, acct_open_date) VALUES (104, 1, 0, 0, '2014-01-15');
```

```
CREATE TABLE holdings_dim(  
  account_id INT, #foreign key from accounts  
  ticker VARCHAR(8), #foreign key and a primary key  
  account_id_ticker VARCHAR(250),  
  value DOUBLE,  
  quantity BIGINT,  
  date DATE,  
  PRIMARY KEY (account_id_ticker)  
);
```

```
INSERT INTO holdings_dim(account_id, ticker, account_id_ticker, value, quantity, date) VALUES (101, 'IXN', '101_IXN', 16625000.00, 114756, '2026-06-15');
```

#16,625,000 / 144.87 = ~114,756 shares

```
INSERT INTO holdings_dim(account_id, ticker, account_id_ticker, value, quantity, date) VALUES (101, 'QQQ', '101_QQQ', 20995000.00, 28219, '2026-06-15');
```

#20,995,000 / 744.00 = ~28,219 shares

```
INSERT INTO holdings_dim(account_id, ticker, account_id_ticker, value, quantity, date) VALUES (101, 'IEF', '101_IEF', 27075000.00, 287185, '2026-06-15');
```

#27,075,000 / 94.28 = ~287,185 shares

```
INSERT INTO holdings_dim(account_id, ticker, account_id_ticker, value, quantity, date) VALUES (101, 'VNQ', '101_VNQ', 8455000.00, 86437, '2026-06-15');
```

#8,455,000 / 97.82 = ~86,437 shares

```
INSERT INTO holdings_dim(account_id, ticker, account_id_ticker, value, quantity, date) VALUES (101, 'GLD', '101_GLD', 21850000.00, 55089, '2026-06-15');
```

#21,850,000 / 396.55 = ~55,089 shares

```
CREATE TABLE security_masterlist(
id BIGINT,
security_name TEXT,
ticker VARCHAR(8),
sec_type VARCHAR(100),
major_asset_class VARCHAR(100),
minor_asset_class VARCHAR(100),
PRIMARY KEY(ticker)
);
```

```
INSERT INTO security_masterlist(id, security_name, ticker, sec_type, major_asset_class, minor_asset_class) VALUES ('1', 'iShares Global Tech ETF', 'IXN', 'ETF', 'Equity', 'Global Technology');
```

```
INSERT INTO security_masterlist(id, security_name, ticker, sec_type, major_asset_class, minor_asset_class) VALUES ('2', 'Invesco QQQ Trust NASDAQ 100', 'QQQ', 'ETF', 'Equity', 'Large Cap Growth');
```

```
INSERT INTO security_masterlist(id, security_name, ticker, sec_type, major_asset_class, minor_asset_class) VALUES ('3', 'iShares 7-10 Year Treasury Bond ETF', 'IEF', 'ETF', 'Fixed Income', 'US Treasury Bonds');
```

```
INSERT INTO security_masterlist(id, security_name, ticker, sec_type, major_asset_class, minor_asset_class) VALUES ('4', 'Vanguard Real Estate ETF', 'VNO', 'ETF', 'Real Assets', 'Real Estate Investment Trust');
```

```
INSERT INTO security_masterlist(id, security_name, ticker, sec_type, major_asset_class, minor_asset_class) VALUES ('5', 'SPDR Gold Shares', 'GLD', 'ETF', 'Commodities', 'Precious Metals');
```

#TABLE WITH PRICES - ONLY ADJUSTED CLOSE PRICES

```
CREATE TABLE pricing_daily_new(
date DATE NOT NULL,
ticker VARCHAR(4) NOT NULL,
ticker_date VARCHAR(15) NOT NULL PRIMARY KEY,
price_type VARCHAR(9) NOT NULL,
price NUMERIC(10,6) NOT NULL
);
```

INSERT INTO WITH PRICE DATA → ONLY ADJUSTED CLOSE TO AVOID SYSTEM CRASH

6337 • `SELECT * FROM pricing_daily_new LIMIT 100;`

	date	ticker	ticker_date	price_type	price
▶	2023-06-16	AVDV	AVDV_2023-06-16	Adj Close	52.650036
	2023-06-20	AVDV	AVDV_2023-06-20	Adj Close	51.835598
	2023-06-21	AVDV	AVDV_2023-06-21	Adj Close	52.105579
	2023-06-22	AVDV	AVDV_2023-06-22	Adj Close	51.592621
	2023-06-23	AVDV	AVDV_2023-06-23	Adj Close	50.656700
	2023-06-26	AVDV	AVDV_2023-06-26	Adj Close	50.764690
	2023-06-27	AVDV	AVDV_2023-06-27	Adj Close	51.016670
	2023-06-28	AVDV	AVDV_2023-06-28	Adj Close	51.187656
	2023-06-29	AVDV	AVDV_2023-06-29	Adj Close	51.133663
	2023-06-30	AVDV	AVDV_2023-06-30	Adj Close	51.799606
	2023-07-03	AVDV	AVDV_2023-07-03	Adj Close	52.123573
	2023-07-05	AVDV	AVDV_2023-07-05	Adj Close	51.808601

pricing_daily_new 39 x Apply

#QUESTION 1 - Return per security and for the whole portfolio

DROP PROCEDURE IF EXISTS q1_returns;

DELIMITER \$\$

CREATE PROCEDURE q1_returns(IN n_months INT)

BEGIN

DECLARE start_date DATE;

DECLARE end_date DATE;

SELECT MAX(date) INTO end_date FROM pricing_daily_new; # most recent day

SET start_date = DATE_SUB(end_date, INTERVAL n_months MONTH);

Total return for each security over the window

SELECT DISTINCT y.ticker, CONCAT(n_months, 'M') AS time_window, ROUND(y.end_price / y.start_price - 1, 4) AS period_return

FROM (SELECT ticker,

FIRST_VALUE(price) OVER (PARTITION BY ticker ORDER BY date) AS start_price,

FIRST_VALUE(price) OVER (PARTITION BY ticker ORDER BY date DESC) AS end_price

FROM pricing_daily_new

WHERE price_type = 'Adj Close'

AND date BETWEEN start_date AND end_date

AND ticker IN ('IXN', 'QQQ', 'IEF', 'VNQ', 'GLD')) y

ORDER BY period_return DESC;

Total return for the whole portfolio over the window (weighted sum)

SELECT 'CLIENT PORTFOLIO' AS portfolio, CONCAT(n_months, 'M') AS time_window,

ROUND(SUM(y.weight * (y.end_price / y.start_price - 1)), 4) AS period_return

FROM (SELECT DISTINCT ticker,

FIRST_VALUE(price) OVER (PARTITION BY ticker ORDER BY date) AS start_price,

FIRST_VALUE(price) OVER (PARTITION BY ticker ORDER BY date DESC) AS end_price,

CASE ticker

WHEN 'IXN' THEN 0.175 WHEN 'QQQ' THEN 0.221

WHEN 'IEF' THEN 0.285 WHEN 'VNQ' THEN 0.089

WHEN 'GLD' THEN 0.230 END AS weight

FROM pricing_daily_new

WHERE price_type = 'Adj Close'

AND date BETWEEN start_date AND end_date

AND ticker IN ('IXN', 'QQQ', 'IEF', 'VNQ', 'GLD')) y;

END\$\$

DELIMITER ;

CALL q1_returns(12);

CALL q1_returns(18);

CALL q1_returns(24);

#QUESTION 2 - CORRELATIONS

DROP PROCEDURE IF EXISTS q2_correlations;

DELIMITER \$\$

CREATE PROCEDURE q2_correlations(IN n_months INT)

BEGIN

DECLARE start_date DATE;

DECLARE end_date DATE;

SELECT MAX(date) INTO end_date FROM pricing_daily_new; # most recent day

SET start_date = DATE_SUB(end_date, INTERVAL n_months MONTH);

SELECT CONCAT(n_months, 'M') AS time_window,

a.ticker AS asset_1,

b.ticker AS asset_2,

ROUND((AVG(a.daily_ret * b.daily_ret) - AVG(a.daily_ret) * AVG(b.daily_ret)) / (STD(a.daily_ret) * STD(b.daily_ret)), 3) AS correlation

FROM (SELECT ticker, date,

(price / LAG(price, 1) OVER (PARTITION BY ticker ORDER BY date)) - 1 AS daily_ret

FROM pricing_daily_new

WHERE price_type = 'Adj Close'

AND date BETWEEN start_date AND end_date

AND ticker IN ('IXN', 'QQQ', 'IEF', 'VNQ', 'GLD')) a

JOIN (SELECT ticker, date,

(price / LAG(price, 1) OVER (PARTITION BY ticker ORDER BY date)) - 1 AS daily_ret

FROM pricing_daily_new

WHERE price_type = 'Adj Close'

AND date BETWEEN start_date AND end_date

AND ticker IN ('IXN', 'QQQ', 'IEF', 'VNQ', 'GLD')) b ON a.date = b.date

WHERE a.daily_ret IS NOT NULL

AND b.daily_ret IS NOT NULL

AND a.ticker < b.ticker

GROUP BY a.ticker, b.ticker

ORDER BY correlation DESC;

END\$\$

DELIMITER ;

CALL q2_correlations(6);

CALL q2_correlations(12);

CALL q2_correlations(18);

CALL q2_correlations(24);

#QUESTION 3 - SIGMA, VOLATILITY - last 6 and 12 months as required in the assignment

DROP PROCEDURE IF EXISTS q3_risk;

DELIMITER \$\$

CREATE PROCEDURE q3_risk(IN n_months INT)

BEGIN

DECLARE start_date DATE;

```

DECLARE end_date DATE;
SELECT MAX(date) INTO end_date FROM pricing_daily_new; # most recent day
SET start_date = DATE_SUB(end_date, INTERVAL n_months MONTH);

# Sigma for each security
SELECT CONCAT(n_months, 'M') AS time_window,
y.ticker,
ROUND(STD(y.daily_ret) * SQRT(250), 4) AS ann_vol
FROM (SELECT ticker, date,
(price / LAG(price, 1) OVER (PARTITION BY ticker ORDER BY date)) - 1 AS daily_ret
FROM pricing_daily_new
WHERE price_type = 'Adj Close'
AND date BETWEEN start_date AND end_date
AND ticker IN ('IXN', 'QQQ', 'IEF', 'VNQ', 'GLD')) y
WHERE y.daily_ret IS NOT NULL
GROUP BY y.ticker
ORDER BY ann_vol DESC;

# QUESTION 3 -Sigma for the whole portfolio (captures diversification)
SELECT CONCAT(n_months, 'M') AS time_window,
'CLIENT PORTFOLIO' AS portfolio,
ROUND(STD(port_ret) * SQRT(250), 4) AS ann_vol
FROM (SELECT date, SUM(CASE
WHEN ticker = 'IXN' THEN daily_ret * 0.175
WHEN ticker = 'QQQ' THEN daily_ret * 0.221
WHEN ticker = 'IEF' THEN daily_ret * 0.285
WHEN ticker = 'VNQ' THEN daily_ret * 0.089
WHEN ticker = 'GLD' THEN daily_ret * 0.230 END) AS port_ret
FROM (SELECT ticker, date,
(price / LAG(price, 1) OVER (PARTITION BY ticker ORDER BY date)) - 1 AS daily_ret
FROM pricing_daily_new
WHERE price_type = 'Adj Close'
AND date BETWEEN start_date AND end_date
AND ticker IN ('IXN', 'QQQ', 'IEF', 'VNQ', 'GLD')) d
WHERE daily_ret IS NOT NULL
GROUP BY date) port;
END$$
DELIMITER ;

CALL q3_risk(6);
CALL q3_risk(12);

# SHARPE RATIO - single 3-year window (36M)
# RF = 3.68% (annualised)

```

```

# Risk-free rate: most recent value of the 3-month U.S. Treasury Bill (3.68%).
# Board of Governors of the Federal Reserve System (US). (2026).
# 3-Month Treasury Bill Secondary Market Rate, Discount Basis [DTB3]
# [Data set]. FRED, Federal Reserve Bank of St. Louis. Retrieved June 20,
# 2026, from https://fred.stlouisfed.org/series/DTB3

```

```

DROP PROCEDURE IF EXISTS q_sharpe;
DELIMITER $$
CREATE PROCEDURE q_sharpe(IN n_months INT)
BEGIN
DECLARE start_date DATE;
DECLARE end_date DATE;
SELECT MAX(date) INTO end_date FROM pricing_daily_new;
SET start_date = DATE_SUB(end_date, INTERVAL n_months MONTH);

# Sharpe for each
SELECT CONCAT(n_months, 'M') AS time_window,
y.ticker,
ROUND(AVG(y.daily_ret) * 250, 4) AS ann_return,
ROUND(STD(y.daily_ret) * SQRT(250), 4) AS ann_vol,
# 0.0368 = risk-free rate: most recent 3-month U.S. T-Bill (3.68%); source: FRED, series DTB3
ROUND((AVG(y.daily_ret) * 250 - 0.0368) / (STD(y.daily_ret) * SQRT(250)), 3) AS sharpe_ratio
FROM (SELECT ticker, date,
(price / LAG(price, 1) OVER (PARTITION BY ticker ORDER BY date)) - 1 AS daily_ret
FROM pricing_daily_new
WHERE price_type = 'Adj Close'
AND date BETWEEN start_date AND end_date
AND ticker IN ('IXN', 'QQQ', 'IEF', 'VNQ', 'GLD')) y
WHERE y.daily_ret IS NOT NULL
GROUP BY y.ticker
ORDER BY sharpe_ratio DESC;

```

```

# Sharpe CURRENT portfolio
SELECT CONCAT(n_months, 'M') AS time_window,
'CLIENT PORTFOLIO' AS portfolio,
ROUND(AVG(port_ret) * 250, 4) AS ann_return,
ROUND(STD(port_ret) * SQRT(250), 4) AS ann_vol,
ROUND((AVG(port_ret) * 250 - 0.0368) / (STD(port_ret) * SQRT(250)), 3) AS sharpe_ratio
FROM (SELECT date, SUM(CASE
WHEN ticker = 'IXN' THEN daily_ret * 0.175
WHEN ticker = 'QQQ' THEN daily_ret * 0.221
WHEN ticker = 'IEF' THEN daily_ret * 0.285
WHEN ticker = 'VNQ' THEN daily_ret * 0.089
WHEN ticker = 'GLD' THEN daily_ret * 0.230 END) AS port_ret

```

```

FROM (SELECT ticker, date,
(price / LAG(price, 1) OVER (PARTITION BY ticker ORDER BY date)) - 1 AS daily_ret
FROM pricing_daily_new
WHERE price_type = 'Adj Close'
AND date BETWEEN start_date AND end_date
AND ticker IN ('IXN', 'QQQ', 'IEF', 'VNQ', 'GLD')) d
WHERE daily_ret IS NOT NULL
GROUP BY date) port;
END$$
DELIMITER ;

```

```
CALL q_sharpe(36);
```

```
# QUESTION 4 AND 5 OPTIMIZED PORTFOLIO
```

```
DROP PROCEDURE IF EXISTS new_sharpe;
```

```
DELIMITER $$
```

```
CREATE PROCEDURE new_sharpe(IN n_months INT)
```

```
BEGIN
```

```
DECLARE start_date DATE;
```

```
DECLARE end_date DATE;
```

```
SELECT MAX(date) INTO end_date FROM pricing_daily_new; # most recent
```

```
SET start_date = DATE_SUB(end_date, INTERVAL n_months MONTH);
```

```
# Sharpe for each
```

```
SELECT CONCAT(n_months, 'M') AS time_window,
```

```
y.ticker,
```

```
ROUND(AVG(y.daily_ret) * 250, 4) AS ann_return,
```

```
ROUND(STD(y.daily_ret) * SQRT(250), 4) AS ann_vol,
```

```
# 0.0368 = risk-free rate: most recent 3-month U.S. T-Bill (3.68%); source: FRED, series DTB3
```

```
ROUND((AVG(y.daily_ret) * 250 - 0.0368) / (STD(y.daily_ret) * SQRT(250)), 3) AS sharpe_ratio
```

```
FROM (SELECT ticker, date,
```

```
(price / LAG(price, 1) OVER (PARTITION BY ticker ORDER BY date)) - 1 AS daily_ret
```

```
FROM pricing_daily_new
```

```
WHERE price_type = 'Adj Close'
```

```
AND date BETWEEN start_date AND end_date
```

```
AND ticker IN ('IXN', 'IEF', 'VNQ', 'GLD', 'AVDV', 'VTIP', 'DBC')) y
```

```
WHERE y.daily_ret IS NOT NULL
```

```
GROUP BY y.ticker
```

```
ORDER BY sharpe_ratio DESC;
```

```
# Sharpe del portafoglio (pesi cliente)
```

```
SELECT CONCAT(n_months, 'M') AS time_window,
```

```
'CLIENT PORTFOLIO' AS portfolio,
```

```
ROUND(AVG(port_ret) * 250, 4) AS ann_return,
```

```
ROUND(STD(port_ret) * SQRT(250), 4) AS ann_vol,
```

```

ROUND((AVG(port_ret) * 250 - 0.0368) / (STD(port_ret) * SQRT(250)), 3) AS sharpe_ratio
FROM (SELECT date, SUM(CASE
WHEN ticker = 'IXN' THEN daily_ret * 0.18
WHEN ticker = 'AVDV' THEN daily_ret * 0.22
WHEN ticker = 'IEF' THEN daily_ret * 0.06
WHEN ticker = 'VTIP' THEN daily_ret * 0.24
WHEN ticker = 'VNQ' THEN daily_ret * 0.05
WHEN ticker = 'DBC' THEN daily_ret * 0.07
WHEN ticker = 'GLD' THEN daily_ret * 0.18 END) AS port_ret
FROM (SELECT ticker, date,
(price / LAG(price, 1) OVER (PARTITION BY ticker ORDER BY date)) - 1 AS daily_ret
FROM pricing_daily_new
WHERE price_type = 'Adj Close'
AND date BETWEEN start_date AND end_date
AND ticker IN ('IXN', 'AVDV', 'IEF', 'VNQ', 'GLD', 'VTIP', 'DBC')) d
WHERE daily_ret IS NOT NULL
GROUP BY date) port;
END$$
DELIMITER ;
CALL new_sharpe(36);

```

```
SELECT * FROM pricing_daily_new LIMIT 100;
```

Appendix B - SQL screenshots CALLS

1. Returns

	ticker	time_window	period_return
▶	IXN	12M	0.6669
	QQQ	12M	0.3992
	GLD	12M	0.2719
	VNQ	12M	0.1314
	IEF	12M	0.0418

	portfolio	time_window	period_return
▶	CLIENT PORTFOLIO	12M	0.2911

	ticker	time_window	period_return
▶	IXN	18M	0.6732
	GLD	18M	0.6194
	QQQ	18M	0.3935
	VNQ	18M	0.1094
	IEF	18M	0.0653

	portfolio	time_window	period_return
▶	CLIENT PORTFOLIO	18M	0.3756

	ticker	time_window	period_return
▶	GLD	24M	0.8478
	IXN	24M	0.7416
	QQQ	24M	0.5507
	VNQ	24M	0.2660
	IEF	24M	0.0797

	portfolio	time_window	period_return
▶	CLIENT PORTFOLIO	24M	0.4929

2. Correlations

	time_window	asset_1	asset_2	correlation
▶	6M	IXN	QQQ	0.962
	6M	IEF	VNQ	0.453
	6M	GLD	QQQ	0.4
	6M	GLD	IXN	0.38
	6M	IEF	QQQ	0.375
	6M	IEF	IXN	0.368
	6M	GLD	IEF	0.286
	6M	QQQ	VNQ	0.178
	6M	GLD	VNQ	0.142
	6M	IXN	VNQ	0.122

	time_window	asset_1	asset_2	correlation
▶	12M	IXN	QQQ	0.951
	12M	IEF	VNQ	0.412
	12M	GLD	IXN	0.301
	12M	GLD	QQQ	0.297
	12M	GLD	IEF	0.239
	12M	IEF	QQQ	0.178
	12M	IEF	IXN	0.173
	12M	QQQ	VNQ	0.172
	12M	GLD	VNQ	0.137
	12M	IXN	VNQ	0.087

	time_window	asset_1	asset_2	correlation
▶	18M	IXN	QQQ	0.962
	18M	QQQ	VNQ	0.448
	18M	IXN	VNQ	0.358
	18M	IEF	VNQ	0.309
	18M	GLD	IXN	0.212
	18M	GLD	QQQ	0.178
	18M	GLD	IEF	0.176
	18M	GLD	VNQ	0.124
	18M	IEF	IXN	0.017
	18M	IEF	QQQ	0.01

	time_window	asset_1	asset_2	correlation
▶	24M	IXN	QQQ	0.962
	24M	QQQ	VNQ	0.384
	24M	IEF	VNQ	0.326
	24M	IXN	VNQ	0.31
	24M	GLD	IXN	0.216
	24M	GLD	IEF	0.189
	24M	GLD	QQQ	0.184
	24M	GLD	VNQ	0.143
	24M	IEF	IXN	-0.014
	24M	IEF	QQQ	-0.024

3. Volatility

	time_window	ticker	ann_vol
▶	6M	GLD	0.3387
	6M	IXN	0.2889
	6M	QQQ	0.1972
	6M	VNQ	0.1384
	6M	IEF	0.0482

	time_window	portfolio	ann_vol
▶	12M	CLIENT PORTFOLIO	0.1224

	time_window	ticker	ann_vol
▶	12M	GLD	0.2734
	12M	IXN	0.2414
	12M	QQQ	0.1732
	12M	VNQ	0.1348
	12M	IEF	0.0466

	time_window	portfolio	ann_vol
▶	6M	CLIENT PORTFOLIO	0.1522

4/5. Optimizations

Current

	time_window	ticker	ann_return	ann_vol	sharpe_ratio
►	36M	GLD	0.2811	0.2027	1.205
	36M	IXN	0.3193	0.2428	1.164
	36M	QQQ	0.2605	0.1994	1.122
	36M	VNQ	0.1056	0.1677	0.41
	36M	IEF	0.0291	0.0654	-0.118

	time_window	portfolio	ann_return	ann_vol	sharpe_ratio
►	36M	CLIENT PORTFOLIO	0.1958	0.1163	1.367

Optimized

	time_window	ticker	ann_return	ann_vol	sharpe_ratio
►	36M	AVDV	0.2521	0.1616	1.332
	36M	GLD	0.2811	0.2027	1.205
	36M	IXN	0.3193	0.2428	1.164
	36M	VTIP	0.0508	0.0197	0.713
	36M	DBC	0.1190	0.1622	0.507
	36M	VNQ	0.1056	0.1677	0.41
	36M	IEF	0.0291	0.0654	-0.118

	time_window	portfolio	ann_return	ann_vol	sharpe_ratio
►	36M	CLIENT PORTFOLIO	0.1911	0.1008	1.531